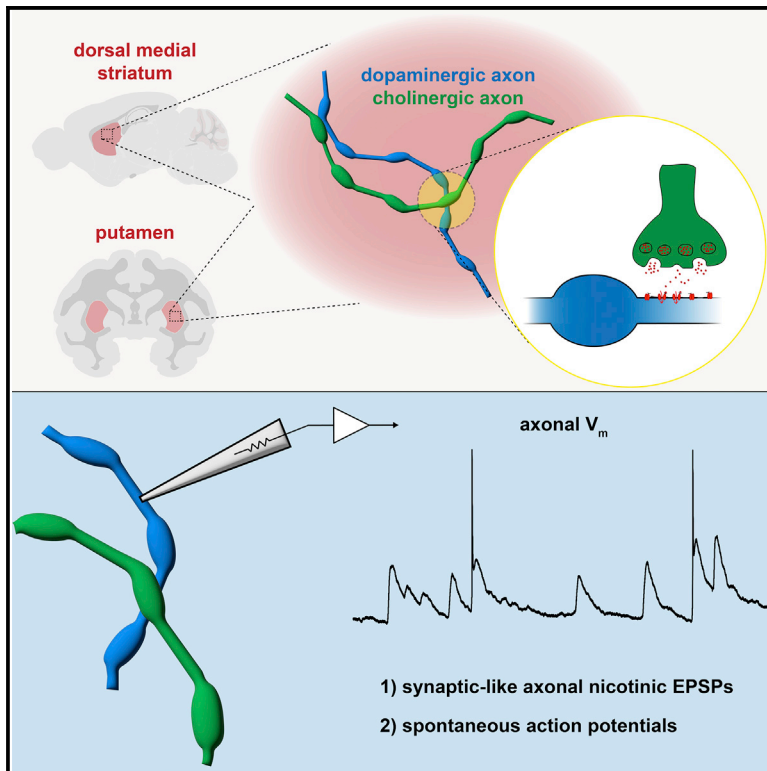


Synaptic-like axo-axonal transmission from striatal cholinergic interneurons onto dopaminergic fibers

Graphical abstract



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In brief

Dopaminergic axons express nicotinic receptors that modulate and evoke neurotransmitter release. Using direct recordings from dopaminergic axons, Kramer et al. reveal the presence of spontaneous synaptic-like nicotinic axonal EPSPs capable of generating spontaneous axonal action potentials. These findings demonstrate the mechanisms underlying nicotinic-receptor-mediated membrane-potential signaling in dopaminergic axons.

Highlights

- Cholinergic interneurons signal to DA axons with phasic axonal EPSPs (axEPSPs)
- AxEPSPs have many characteristics similar to synaptic depolarizations
- Spontaneous axEPSPs can evoke spontaneous action potentials locally in the axon
- AxEPSPs shape input-output function of DA axons, distinct from somatic signaling



Report

Synaptic-like axo-axonal transmission from striatal cholinergic interneurons onto dopaminergic fibers

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SUMMARY

Transmission from striatal cholinergic interneurons (CINs) controls dopamine release through nicotinic acetylcholine receptors (nAChRs) on dopaminergic axons. Anatomical studies suggest that cholinergic terminals signal predominantly through non-synaptic volume transmission. However, the influence of cholinergic transmission on electrical signaling in axons remains unclear. We examined axo-axonal transmission from CINs onto dopaminergic axons using perforated-patch recordings, which revealed rapid spontaneous EPSPs with properties characteristic of fast synapses. Pharmacology showed that axonal EPSPs (axEPSPs) were mediated primarily by high-affinity $\alpha 6$ -containing receptors. Remarkably, axEPSPs triggered spontaneous action potentials, suggesting that these axons perform integration to convert synaptic input into spiking, a function associated with somatodendritic compartments. We investigated the cross-species validity of cholinergic axo-axonal transmission by recording dopaminergic axons in macaque putamen and found similar axEPSPs. Thus, we reveal that synaptic-like neurotransmission underlies cholinergic signaling onto dopaminergic axons, supporting the idea that striatal dopamine release can occur independently of somatic firing to provide distinct signaling.

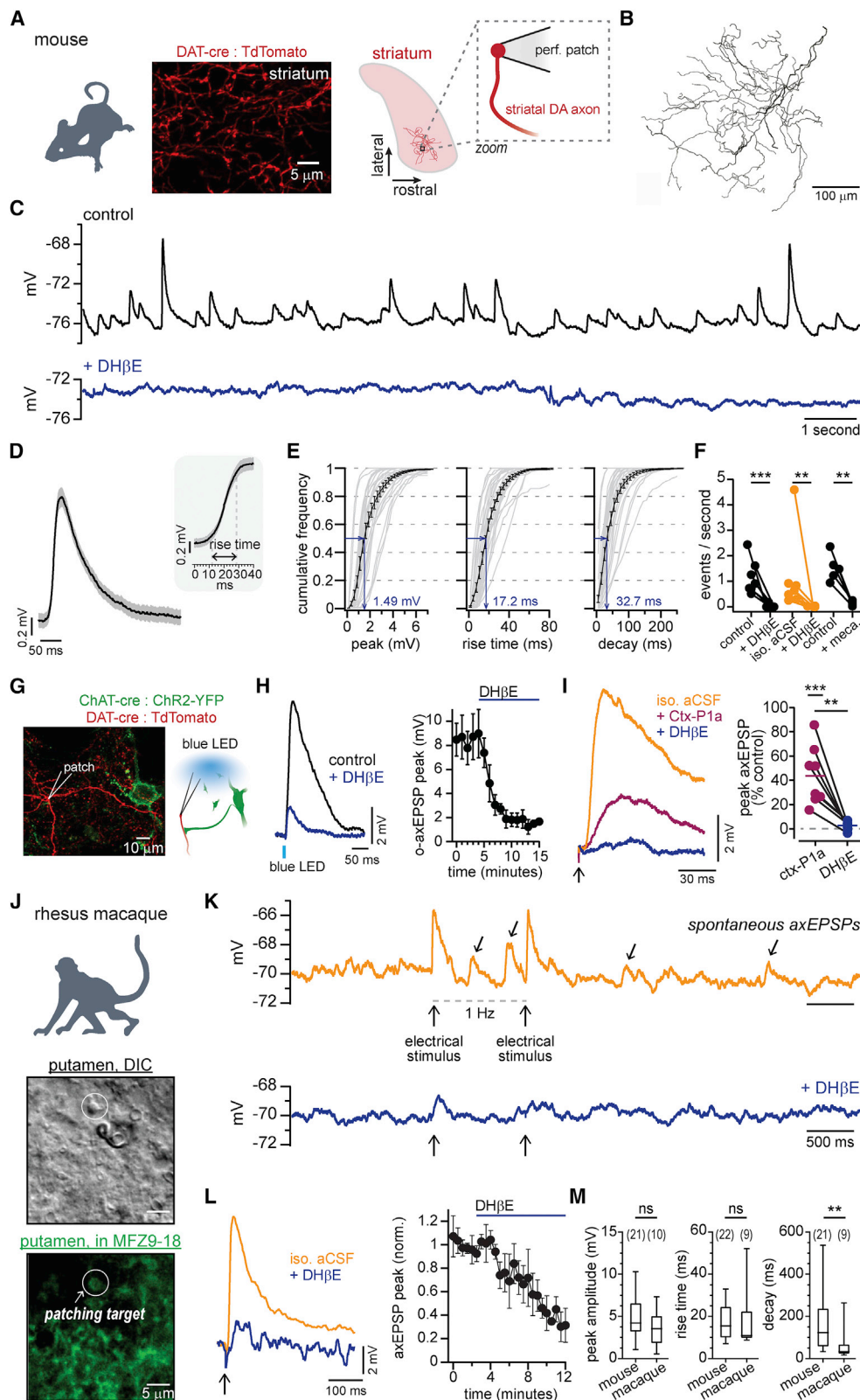
INTRODUCTION

Nicotinic acetylcholine receptors (nAChRs) modulate neurotransmitter release in the nervous system through their actions on presynaptic membranes (McGehee et al., 1995; Gray et al., 1996). At the circuit and cellular levels, presynaptic nAChRs have been shown to promote transmitter release and filter frequency patterns of axonal signals (Role and Berg, 1996; Wonnacott, 1997; Dani and Bertrand, 2007; Threlfell and Cragg, 2011). In behavioral experiments, presynaptic nAChRs are thought to enhance stimulus detection (Disney et al., 2007; Howe et al., 2010) and regulate drug reinforcement (Brunzell et al., 2010). Despite an extensive literature focusing on the function of presynaptic nAChRs in various brain circuits, how these receptors are activated by physiological cholinergic transmission has yet to be determined.

In the striatum, interactions between cholinergic and dopaminergic signaling have been proposed to play roles in reward,

motivation, and motor learning (Aosaki et al., 1994; Ding et al., 2010; Collins et al., 2016; Howe et al., 2019), and their imbalance is thought to contribute to the symptoms of Parkinson's disease (Lim et al., 2014; Tanimura et al., 2019; Tanimura et al., 2018). Previous work has shown that dopamine (DA) release can be modulated by striatal cholinergic interneurons (CINs) through the activation of nAChRs expressed on the axon of midbrain dopaminergic neurons (Zhou et al., 2002; Sulzer et al., 2016). Studies have shown that the activation of CINs using optogenetics (Cachope et al., 2012; Threlfell et al., 2012), stimulation of cortical inputs (Kosillo et al., 2016; Adrover et al., 2020), or spontaneous CIN activity (Zhou et al., 2001; Yorgason et al., 2017) can trigger release directly from dopaminergic axons. Similarly, behavioral work has reported that a mismatch may exist between DA released from striatal projection axons and somatic spiking (Hamid et al., 2016; Mohebi et al., 2019), and it has been suggested that the difference results from local control of DA release by axonal nAChRs (Mohebi and Berke, 2020).





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However, other work reported that DA release is largely represented by somatic spiking (Kim et al., 2020), questioning whether local modulation of these axons in the striatum can produce an independent signal. Therefore, in order to understand the role of axonal nAChR-mediated DA release in behavior, we first need a better understanding of the functional consequences of acetylcholine (ACh) released from CIN axons onto dopaminergic axons.

While nAChR modulation of striatal DA signaling has been explored extensively, most studies have focused on CIN-evoked DA release, which has left a substantial gap in our understanding of how cholinergic transmission directly influences the excitability of dopaminergic axons. In particular, CIN-evoked DA release involves a disynaptic signaling cascade that integrates multiple processes into a single DA measurement. Factors that contribute to this process include calcium-dependent release for both ACh and DA, neurotransmitter diffusion and regulation by transporters and degradation enzymes, as well as the voltage-gated ion channels that regulate axonal excitability. Therefore, the complexities of this disynaptic signaling process have made it challenging to infer the specific contribution of any one factor, such as the effects of nAChR activation on axonal membrane dynamics during ACh transmission.

Current views regarding ACh-DA interactions are based on structural data from electron microscopy (EM) studies. These studies find that CIN terminals lack morphological characteristics that classically define a synapse and thus suggest that transmission likely occurs through slow, volume-based diffusion. For example, axonal nAChRs on dopaminergic fibers have only rarely been observed in close apposition to presynaptic release sites (Contant et al., 1996; Jones et al., 2001; Aznavour et al., 2003). An earlier immunocytochemical EM study examining cholinergic and dopaminergic axons did not identify classic synaptic specializations but found numerous close contacts between a few sets of varicosities (Chang, 1988). Therefore, the anatomy of the CIN-DA axonal contacts is suggestive of vol-

ume transmission; however, a lack of subthreshold recordings from the terminal region of these axons leaves the functional properties of this axo-axonal neurotransmission unexplored.

Here, we examined cholinergic transmission onto axonal nAChRs using direct recordings from striatal dopaminergic axons of mice and primates in brain slices that separate the axon from the soma and dendrites. We found that CIN-mediated activation of nAChRs resulted in fast, phasic depolarizations in the axon that closely resembled nAChR-mediated excitatory postsynaptic potentials (EPSPs) recorded in somas and dendrites of cells in the cortex and thalamus (Bennett et al., 2012; Sun et al., 2013; Obermayer et al., 2019). Our direct axonal recordings show that nAChRs control electrical signaling in dopaminergic axons through a mechanism involving synaptic-like activation of $\alpha 6$ subunit-containing nAChRs. Furthermore, these spontaneous synaptic-like EPSPs can generate spontaneous local action potentials, which could comprise a local striatal mechanism of transmitter release from the dopaminergic axon. We thus provide new insight into the critical role of nAChRs in controlling striatal DA release and establish a mechanistic framework for understanding input-output signaling in dopaminergic axons independently from somatic action potential initiation.

RESULTS

Characteristics of spontaneous nicotinic EPSPs recorded in dopaminergic axons of rodents and primates

Subthreshold voltage dynamics were measured in dopaminergic axons located within the dorsomedial striatum. Axons were identified and visualized by selective tdTomato expression in neurons expressing the DA transporter (Figure 1A). To measure the axonal membrane potential, we performed perforated-patch recordings from the cut ends of axons (blebs) located on the surface of the slice (Shu et al., 2007). Perforated-patch recording was used to prevent the rapid rundown of nicotinic currents, as reported in cell bodies of dopaminergic neurons (Wu et al.,

Figure 1. Spontaneous nicotinic axEPSPs recorded in dopaminergic axons

- TdTomato-expressing axons of midbrain dopaminergic neurons from mice and schematic of perforated patch axonal recording configuration.
- Morphological reconstruction of an axon filled with neurobiotin.
- Spontaneous axEPSPs recorded from an axon in the dorsomedial striatum, control (black) and after DH β E (blue).
- Averaged trace of 334 spontaneous axEPSPs from one axonal recording. Inset indicates rising phase.
- Cumulative histograms for the distribution of axEPSP peak amplitudes, rise time, and decay time.
- Compiled axEPSP frequencies in control and in drug. Black traces recorded in control aCSF, and orange traces recorded in isolation aCSF. Mecamylamine (10 μ M) abbreviated as "meca."
- Image from a DAT-Cre x ChAT-Cre mouse injected with DIO-ChR2 into the dorsal medial striatum and FLEX-tdTomato into the SNc, and a cartoon schematic of the experiment.
- Example traces of axEPSPs evoked by optogenetic activation of ChAT+ terminals recorded in control aCSF and after DH β E (left). Averaged time course showing the inhibition of the optically stimulated axEPSP by DH β E (right).
- Example traces of electrically-stimulated axEPSPs in isolation aCSF, conotoxin P1a (ctx-P1a), and DH β E. Right: peak amplitude as a percentage of each axon's control following ctx-P1a and DH β E.
- Experimental setup for experiments in rhesus macaque. Dopaminergic axons were identified using MFZ9-18. Below: example images of a slice showing labeled axons and corresponding DIC image.
- Example traces of axEPSPs recorded in isolation aCSF and following DH β E from macaque.
- Example stimulated axEPSPs in isolation aCSF and DH β E (left) along with averaged time course (right) in 4 axons.
- Properties of stimulated axEPSPs recorded in macaque compared with those from the same conditions recorded in mice.

** $p < 0.001$; *** $p < 0.0001$; ns, $p > 0.05$.

Data in (E), (I), and (L) are represented as mean $\pm$ SEM. Box plots show medians and 25th/75th quartiles, and whiskers are maximum and minimum values.

See also Figure S2.

2004). In some experiments, the patch was ruptured to allow the filling of the axonal cytoplasm with neurobiotin. Post-hoc visualization confirmed that the recorded axons were terminal arbors that had been isolated from the main axon trunk during slice preparation (Figure 1B).

Our recordings revealed the presence of EPSP-like depolarizations of the subthreshold membrane potential of dopaminergic axons. As shown in the Figure 1C example, the axEPSPs were small amplitude, fast, phasic events. Across axonal recordings, axEPSPs occurred at a median of 51.2 events per minute, had a median amplitude of 1.49 mV, a median rise time of 17.2 ms, and a median decay time of 32.7 ms ($n = 28$; Figures 1D and 1E). The axEPSPs were completely abolished by dihydro- β -erythroidine hydrobromide (DH β E, 1 μ M), an nAChR antagonist with modest selectivity for receptors containing the $\alpha 4\beta 2$ subunits (Harvey et al., 1996), and by the broad nicotinic antagonist mecamylamine (10 μ M; Figures 1F and S2A). We saw similar results when we recorded in artificial cerebrospinal fluid (aCSF) containing CNQX, AP-5, CGP, sulpiride, atropine, and gabazine to block AMPA, NMDA, GABA-B, dopamine D2, muscarinic, and GABA-A receptors to isolate ACh transmission (isolation aCSF) (Figures 1F and S2B). Furthermore, there was no difference in the frequency of events observed between male and female mice (Figure S2C). We therefore find spontaneous nicotinic axEPSPs in the striatal axon of dopaminergic neurons. By contrast, axEPSPs were never observed in recordings from the main axon trunk (Kramer et al., 2020), supporting the conclusion that axEPSPs are generated by input to the axonal terminals.

We next directly tested whether the axEPSPs originated from CIN-mediated transmission. To test this, we crossed choline acetyltransferase (ChAT)-Cre mice with dopamine transporter (DAT)-Cre mouse lines (ChAT-Cre/DAT-Cre) and expressed channelrhodopsin (ChR2) and tdTomato in CINs and midbrain dopaminergic neurons, respectively (Figure 1G). Light stimulation of the CINs resulted in optically evoked phasic depolarizations (Figure 1H). With the LED power tuned below the threshold of evoking an axonal action potential (AP), we found that optically evoked axEPSPs had a mean amplitude of 8.5 ± 1.66 mV and were inhibited by DH β E ($n = 6$). We also measured electrically stimulated cholinergic transmission using a bipolar electrode placed in the dorsomedial striatum and recorded in cholinergic isolation aCSF. Electrically evoked axEPSPs tuned below the threshold of evoking an axonal AP had a mean amplitude of 5.65 ± 0.76 mV with a coefficient of variation (CV) of 0.16 ± 0.03 ($n = 16$) and were also blocked by DH β E (Figure S2F). Importantly, the kinetics and latency to onset of the electrically evoked axEPSPs did not differ in control versus cholinergic isolation solutions, consistent with the direct release of ACh onto dopaminergic axons (Figures S2G–S2K).

$\alpha 6$ subunit-containing nAChRs are expressed on the axons of dopaminergic neurons (Zoli et al., 2002), where they contribute to nAChR-dependent DA release in the dorsal striatum (Exley et al., 2008). Yet, the functional effect of this subunit on subthreshold nicotinic depolarization in the axon is unknown. We therefore tested the effect of blocking $\alpha 6$ subunit-containing nAChRs on the stimulated axEPSP by using conotoxin-P1a (ctx-P1a), an antagonist that has a high selectivity for $\alpha 6$ subunit-containing

nAChRs (Dowell et al., 2003; McIntosh et al., 2004). Ctx-P1a (1 μ M) reduced the amplitude of the electrically stimulated axEPSP to $43.8 \pm 8.48\%$ of the control amplitude (Figure 1I; $t(7) = 6.63$, $p = 0.0006$, $n = 8$) and significantly slowed the axEPSP rise time (Figure S2L; control: 14.4 ± 3.4 ms; ctx-P1a: 23.5 ± 2.7 ms; $n = 8$). Examining the effects of ctx-P1a on spontaneous axEPSPs (Figures S2D and S2E), we found that the remaining events in ctx-P1a had smaller amplitudes in the upper quartile of events and longer rise times overall (Figure S2E), similar to evoked axEPSPs following ctx-P1a treatment. These results show that $\alpha 6$ -containing receptors account for the majority of the large amplitude spontaneous nicotinic axEPSPs in dopaminergic axons.

To test the generalizability of our observations across species, we asked whether cholinergic transmission produced synaptic-like depolarizations in dopaminergic axons projecting to the putamen nucleus of non-human primates. To this end, we performed perforated-patch clamp recordings from dopaminergic axons of rhesus macaques (Figure 1J), which were visualized by incubating tissue slices in a fluorescently labeled cocaine analog MFZ9-18 (Eriksen et al., 2009). We observed spontaneous and evoked axEPSPs (Figure 1K), and both were inhibited by DH β E (Figures 1L and S2O). We found the stimulated axEPSPs were similar in amplitude, kinetics, and short-term plasticity characteristics to those recorded in mice. Interestingly, the median decay was significantly faster in macaques (62.4 ± 26.1 ms, $n = 9$) than in mice (168 ± 33.3 ms, $n = 21$; $p = 0.004$; Figures 1M and S2N). We also found that stimulated axEPSPs in macaque axons had a similar axEPSP amplitude CV (0.23 ± 0.5 , $n = 11$) to those recorded in mice. Altogether, these results provide evidence that the synaptic-like depolarizations recorded in mice are also present in the dopaminergic axons of primates.

Evidence for synaptic-like transmission of ACh onto dopaminergic axons

At conventional synapses, spontaneous fusion of a single neurotransmitter vesicle can be recorded as a miniature postsynaptic event in the absence of AP-driven release. We were therefore interested in whether the spontaneous axEPSPs persist in the presence of tetrodotoxin (TTX) to block sodium dependent APs, which could be indicative of a synaptic connection. Bath application of TTX led to a ~ 5 -fold reduction in the frequency of spontaneous axEPSPs. Importantly, we observed miniature nicotinic axEPSPs that persisted in the presence of TTX and were blocked by the further addition of DH β E (Figures 2A–2C). The rise time of axEPSPs was similar between TTX and control conditions (ctrl: 16.8 ± 0.38 ms, TTX: 16.2 ± 0.83 ms; $n = 9$, $p = 0.08$), while the amplitude was reduced (ctrl: 1.59 ± 0.02 mV; TTX: 1.31 ± 0.04 mV; $p < 0.0001$; Figure 2D), predominantly affecting large amplitude events (Figure 2E). Interestingly, the decay kinetics of the miniature events were substantially faster in the presence of TTX (33.4 ± 0.70 in control to 26.6 ± 0.92 ms in TTX, $p < 0.0001$), suggesting that TTX-sensitive sodium channels likely boost and prolong the decay of axEPSPs. Thus, we observe the presence of miniature axEPSPs in dopaminergic axons, which may indicate a close apposition between a set of axonal nAChRs and ACh vesicular release sites.

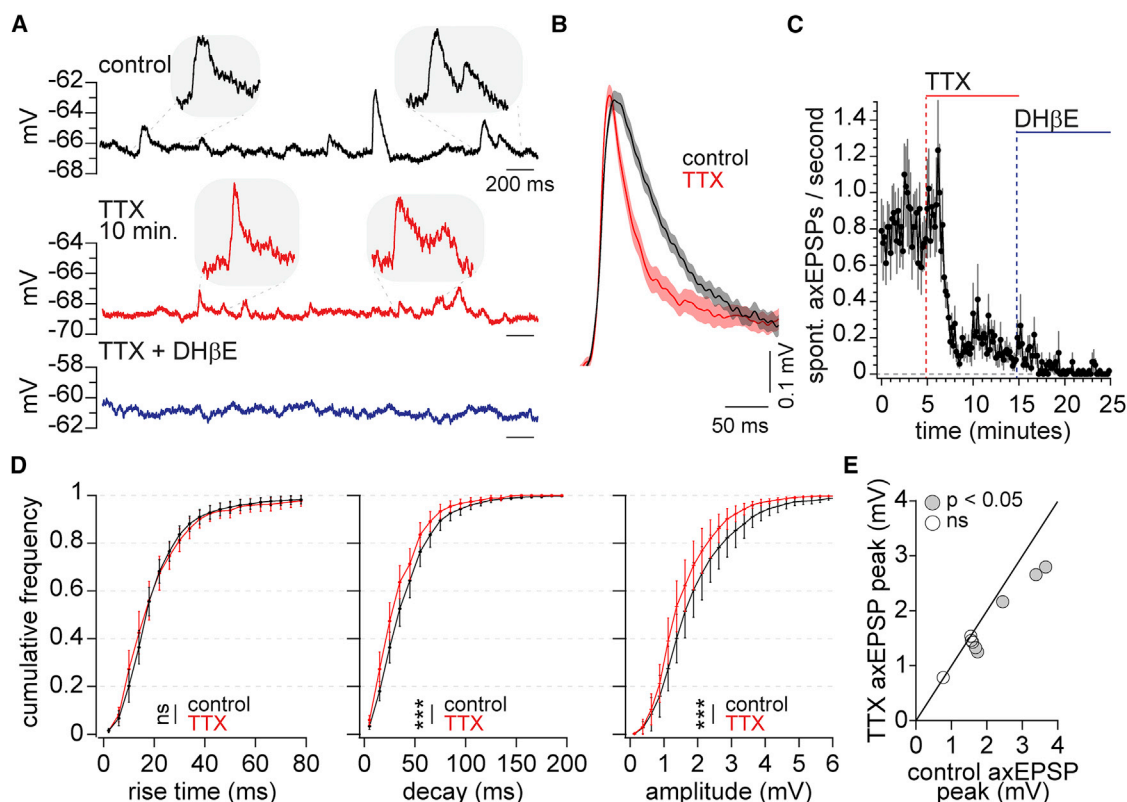


Figure 2. Spontaneous "miniature" nicotinic axEPSPs recorded from dopaminergic axons

(A) Spontaneous axEPSPs measured in control, in the presence of TTX (500 nM), and TTX plus DHβE (1 μM). Example traces are from the same axon.

(B) Average of axEPSPs from one axon in control (358 events) and in TTX (176 events).

(C) Averaged time course of spontaneous axEPSP frequency following TTX and DHβE in 9 axons.

(D) Cumulative histograms comparing control and TTX axEPSPs in rise time, decay, and peak amplitude.

(E) Average axEPSP event amplitudes in control plotted against average in TTX. Each point is the average from all control or TTX event amplitudes plotted against each other from one recording. Solid line is a line of unity.

Data in (C) and (D) are represented as mean ± SEM.

We next tested the extent to which axo-axonal transmission between CINs and dopaminergic axons is regulated by acetylcholinesterase (AChE). The striatum has among the highest expressions of AChE of anywhere in the brain (Hoover et al., 1978; Zhou et al., 2001). In other peripheral and central regions, AChE is localized to synapses where it is thought to limit free diffusion of ACh outside of the synapse following release events. To probe for possible receptors outside a synaptic contact, we used ambenonium to inhibit AChE, hypothesizing that this would have a predominant effect on receptors distal to release sites with a longer diffusion distance (Szapiro and Barbour, 2007; Bennett et al., 2012).

The kinetics of spontaneous axEPSPs were altered following the inhibition of AChE. AxEPSPs across axons had substantially slowed decay kinetics in ambenonium, as reflected by a larger average area under the curve (AUC) and longer rise times (Figures 3E and 3F; ctrl rise time: 20.8 ± 0.23 ms versus amb-early: 24.1 ± 0.28 ms, $n = 10$, $p = 0.003$; ctrl versus amb-late: 26.5 ± 0.33 ms, $p < 0.0001$). Inhibition of AChE also resulted in a brief increase in the frequency of spontaneous axEPSPs ($159 \pm 16.4\%$ of control) followed by a long-lasting decrease

($87.6 \pm 12.8\%$ of control; Figures 3C and 3D). These results are consistent with the idea that AChE inhibition produces longer lasting ACh signaling and broader ACh diffusion beyond CIN terminal release sites.

Similar to the analysis of volume transmission in other systems (Szapiro and Barbour, 2007; Bennett et al., 2012; Coddington et al., 2013), we reasoned that if diffusion of ACh across long distances is required to activate the nAChRs on dopaminergic axons, then the inhibition of AChE should result in the potentiation of the axEPSP amplitude. Following inhibition of AChE, however, we found that ambenonium did not potentiate the peak amplitude of stimulated (Figures 3A and 3B; ctrl: 4.8 ± 0.7 mV versus amb-early: 4.7 ± 0.7 mV, $p > 0.99$; ctrl versus amb-late: 3.0 ± 0.3 mV, $n = 8$, $p = 0.042$) or spontaneous axEPSP amplitude (Figures 3E and 3F; ctrl: 1.00 ± 0.01 mV versus amb-early: 1.02 ± 0.02 mV, $n = 10$, $p = 0.228$; ctrl versus amb-late: 0.87 ± 0.01 mV, $p = 0.005$). Rather, there was a significant reduction in the peak amplitudes during the late phase of ambenonium wash-in, likely reflecting receptor desensitization due to an increased concentration of extracellular ACh. Because CINs express presynaptic muscarinic autoreceptors that inhibit release,

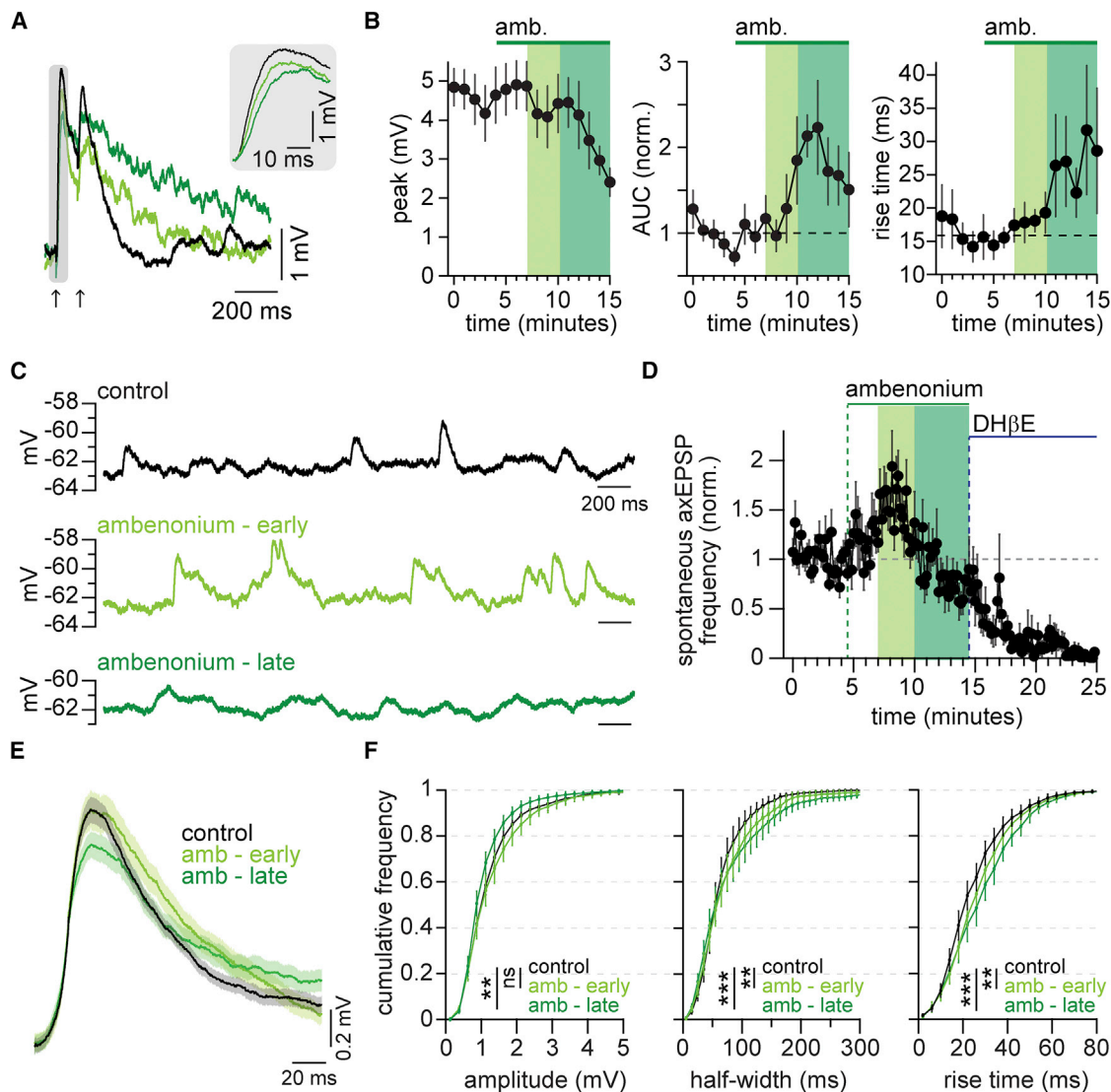


Figure 3. Kinetics of axEPSPs, but not amplitude, are constrained by AChE

(A) Example stimulated axEPSPs in control and in ambenonium (arrows indicate stimulation times). Inset shows an enhanced view of the rising phase.

(B) Stimulated axEPSPs were analyzed for the effect of ambenonium on peak amplitude, area under the curve (AUC), and rise time.

(C) Example traces of axEPSPs in control, in the early phase of ambenonium wash-in, and in the late phase of ambenonium.

(D) Time course of spontaneous axEPSP frequency (normalized) following the application of ambenonium, followed by DHβE.

(E) Example spontaneous event averages from a single axonal recording in control and in ambenonium.

(F) Cumulative histograms of axEPSP amplitude, half-width, and rise time in control and ambenonium.

** $p < 0.01$; *** $p < 0.001$; ns, $p > 0.05$.

Data in (B), (D), and (F) are represented as mean $\pm$ SEM.

See also Figure S3.

one possibility is that the reduction in peak axEPSP may result from autoreceptor inhibition caused by the increase in ambient ACh. However, recordings made in atropine to inhibit muscarinic receptors produced similar results (Figures S3C–S3F). Furthermore, there was no change in the paired-pulse ratio (PPR) during ambenonium (Figure S3B). These results demonstrate that pre-synaptic inhibition from muscarinic autoreceptors does not account for the lack of peak potentiation of axEPSPs in ambeno-

nium. Together, these data show that AChE regulates the spatial and temporal release of ACh onto dopaminergic axons in the striatum. Unexpectedly, however, we found that the inhibition of AChE does not result in potentiation of axEPSPs, a finding that is inconsistent with classical notions of volume transmission. These results therefore suggest an arrangement whereby a fraction of CIN terminals exist in close proximity to nAChRs on dopaminergic axons.

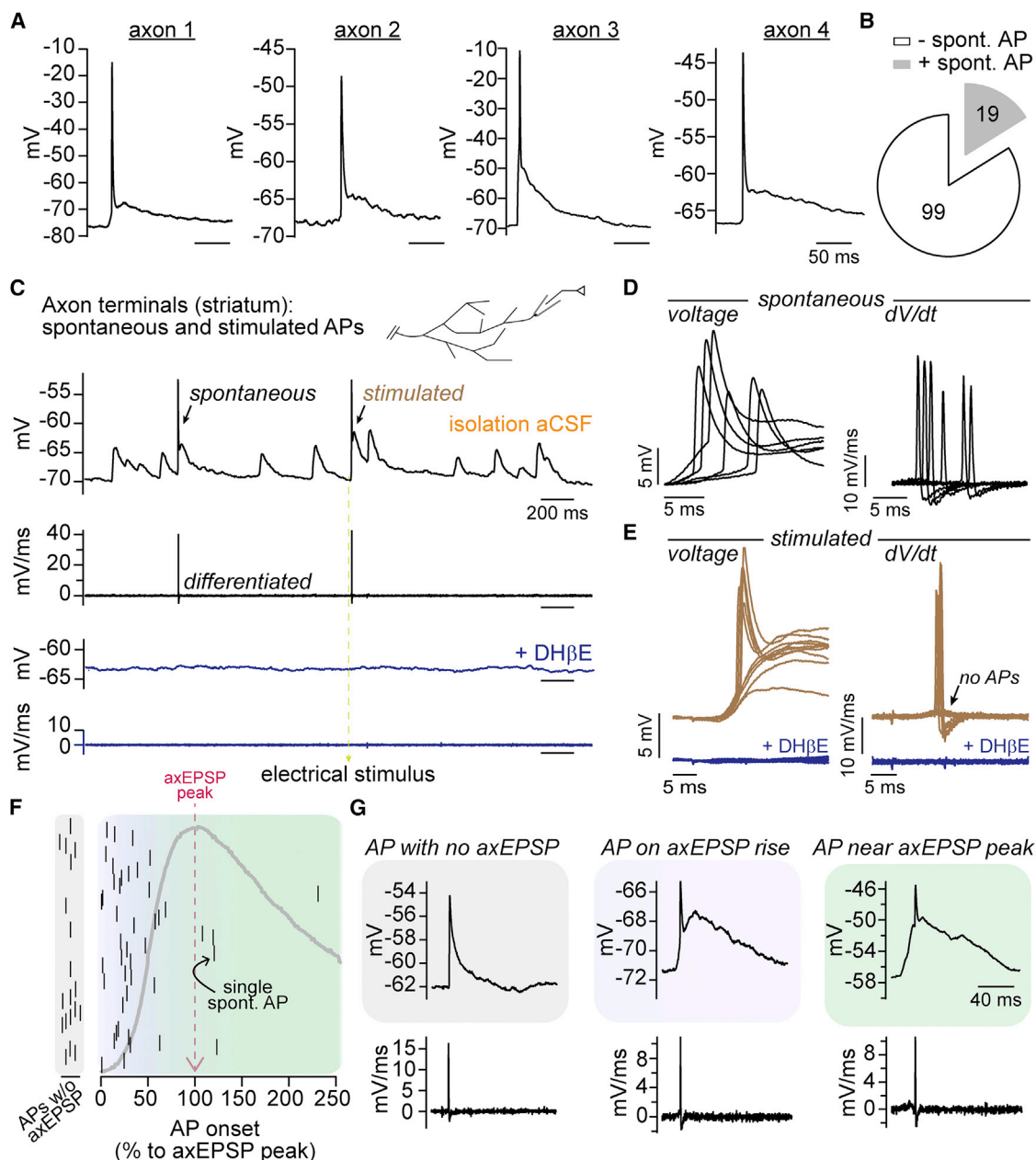


Figure 4. Spontaneous action potentials in dopaminergic axons triggered by nicotinic axEPSPs

(A) Spontaneous APs recorded in four separate dopaminergic axons.

(B) Pie chart of the number of axonal recordings that did (gray) or did not (white) contain at least one identified spontaneous AP.

(C) Example axonal recording made in isolation solution. Left arrow points to a spontaneous AP; right arrow points to a stimulated AP; below is the differentiated trace. Blue traces are from the same time window as the control traces after DH β E application, showing voltage (top) and differentiated trace (bottom).

(D) All spontaneous APs from the same recording shown in (C), aligned to the onset of the axEPSP.

(E) Stimulated axEPSPs and stimulated APs from same axonal recording shown in (C), aligned to 4 ms before stimulation; blue shows that stimulated APs are abolished after DH β E.

(F) Plot of AP timing relative to the underlying axEPSP (APs with no axEPSP are separated to the left). The y axis is an arbitrary count. An example axEPSP is shown in gray for reference, with its peak aligned to the x axis.

(G) Examples of spontaneous APs in three different scenarios: AP with no underlying axEPSP, AP occurring on the rising phase of the axEPSP, or AP occurring near the axEPSP peak.

See also Figure S4.

AxEPSPs contribute to the generation of spontaneous action potentials in dopaminergic axons

Studies from a variety of central neurons have proposed that the primary function of presynaptic nAChRs is to modulate neurotransmitter release through many possible mechanisms (Wonnacott, 1997; MacDermott et al., 1999; Dani and Bertrand, 2007). In the present study, we find that nAChR signaling in dopaminergic axons by CINs likely modulates DA release by shaping the subthreshold voltage. Yet, the presence of axEPSPs in these axons raises the question of whether the summation of axEPSPs may directly trigger APs in the axon terminals, independently of conventional somatic input and integration.

We were surprised to find that a subset of axonal recordings (19/118 axons) exhibited nicotinic axEPSPs that were accompanied by spontaneous axonal APs (Figures 4A and 4B). Spontaneous APs could be identified as a characteristically abrupt increase in the first derivative of the membrane potential (dV/dt), which was also seen in all-or-none APs evoked from the same axons (Figures 4C–4E). Because perforated-patch recordings with glass pipettes result in the filtering of axonal APs and a reduction in peak amplitude (Ritzau-Jost et al., 2021; Oláh et al., 2021), we compared spontaneous axonal APs recorded from the main trunk of the dopaminergic axon in perforated-patch recordings and constructed a weighted Z score to identify spontaneous APs in striatal perforated-patch recordings (Figure S4). Using this method, we identified 63 spontaneous axonal APs from 19 axonal recordings. We found that in 45 APs, the occurrence of the axonal spike overlapped with the depolarization from an axEPSP, with most spikes occurring during the rising phase (Figures 4F and 4G). The remaining 18 APs occurred in the absence of any resolvable axEPSP, suggesting active propagation of the AP past the point where the subthreshold axEPSP had passively decayed. These data strongly suggest that spontaneous nicotinic release events that produce axEPSPs can trigger spontaneous all-or-none APs in the terminal region of dopaminergic axons.

DISCUSSION

In this study, we examine the mechanisms of cholinergic transmission onto dopaminergic axons and their control of axonal excitability. Our results demonstrate that cholinergic transmission onto dopaminergic axons has some non-synaptic properties but, surprisingly, also exhibits several functional hallmarks of signaling observed at standard point-to-point synapses. In addition, our direct axonal recordings show that nAChR activation leads to the generation of spontaneous action potentials. Together, these results provide a mechanistic framework for understanding signal processing in dopaminergic axons that occurs independently from somatic action potential initiation.

Synaptic and non-synaptic functional features of ACh release onto dopaminergic axons

Previous conceptualizations of cholinergic transmission onto dopaminergic axons were based solely on structural studies which predicted that presynaptic nAChRs are likely activated by diffuse volume transmission (Descarries et al., 1997). In contrast to the slow signaling associated with volume transmis-

sion, our functional recordings suggest that cholinergic transmission occurs over a range of signaling speeds that includes rapid signaling through axonal nAChRs. Quantification of spontaneous and evoked axEPSPs revealed a distribution of fast rise times, with the median value being at ~ 17 ms and a relatively wide standard deviation to include a fast lower quartile ranging from 5.56 to 10.3 ms and slow upper quartile ranging from 26.1 to 60.5 ms. These values are roughly comparable to those reported for fast cholinergic transmission onto somatodendritic postsynaptic sites in cells from the retina, cortex, and hippocampus (Bennett et al., 2012; Sun et al., 2013; Obermayer et al., 2019; Sethuramanujam et al., 2021). It is important to consider that the kinetic values reported here are likely influenced by the perforated-patch recording conditions and by filtering due to axonal cable properties. These limitations make the axEPSP rise times reported in our study likely an underestimate of the true value.

Our results show that axEPSPs exhibit many functional characteristics that are typically observed in conventional synaptic transmission. For example, following the blockade of AP-dependent transmission with TTX, we observed miniature axEPSPs. We also observed that the peak amplitude of electrically evoked axEPSPs was insensitive to AChE inhibition by ambenonium. These observations suggest that there is a population of nAChRs on dopaminergic axons closely localized near ACh release sites. Other characteristics of this cholinergic transmission are more consistent with non-synaptic transmission. For example, the peak amplitude of stimulated axEPSP shows a relatively small CV of ~ 0.16 , a value which is comparable to non-synaptic glutamate transmission observed in the cerebellum and cortex (Szapiro and Barbour, 2007; Bennett et al., 2012). Furthermore, the latency to the onset of axEPSPs is ~ 6 ms on average, which is slow compared with conventional synaptic transmission for glutamate (Geiger et al., 1997; Szapiro and Barbour, 2007) and ACh (Hefft et al., 1999). However, this value aligns closely with an estimate for ACh transmission onto dopaminergic axons of 7 ms extrapolated from carbon-fiber recordings of DA release (Wang et al., 2014).

We find that cholinergic transmission onto dopaminergic axons exhibits functional similarities to synaptic signaling, but it also has features of signaling at non-synaptic sites. This form of hybrid neurotransmission, with both synaptic and non-synaptic properties, has recently been proposed to underlie cholinergic signaling throughout the brain (Disney and Higley, 2020) and has been described empirically in the retina (Sethuramanujam et al., 2021). In light of past ultrastructural studies within the striatum and our functional data, therefore, we propose a similar model of transmission that rejects a dichotomous conceptualization of neurotransmission as either synaptic or non-synaptic in favor of a more nuanced continuum of transmission modalities.

Toward understanding the biophysics of axo-axonal CIN to DA transmission

Our functional data are compatible with a range of axo-axonal contact architectures that likely includes ACh release sites closely apposed to dopaminergic axons and others that are further apart. For example, we observed a low frequency of

miniature axEPSPs, events classically assumed to result largely from release sites that form point-to-point contacts. Of note, this type of contact has been described morphologically in at least a few instances for cholinergic to dopaminergic axons (Chang, 1988). As a nonexclusive alternative, axEPSPs may occur in the absence of classical synaptic contacts, resulting instead from the characteristics of axonal nAChRs together with properties intrinsic to the axon. Dopaminergic axons express $\alpha 6$ subunit-containing nAChRs, which have a high affinity for ACh with an EC_{50} of ~ 100 nM (Salminen et al., 2007; Drenan et al., 2008; Grady et al., 2010; Chen et al., 2018). In addition, these axons have high average input resistances of ~ 1.8 G Ω (Kramer et al., 2020). Together, these characteristics could allow for large nAChR signals by the diffuse activation of a few receptors. Similar results have been observed in the retina, where currents recorded in the ganglion cells can resemble synaptically generated currents, even following the diffusion of ACh from release sites located tens of nanometers and up to a micrometer away (Sethuramanujam et al., 2021).

When we inhibited AChE to remove it as an ACh diffusion barrier, we found an unexpected transient increase in the frequency of detectable spontaneous axEPSPs. This increase in frequency coincided with a slowing of the rise and decay constants during the early phase of ambenonium application. These data could be consistent with the activation of nAChRs by a more diffuse concentration of ACh at newly accessible nAChR sites previously inaccessible due to the action of AChE. An alternative hypothesis is that there was a selective increase in the small-amplitude axEPSPs that were previously undetectable in the baseline noise of the trace, which would appear as an increase in frequency. The late decrease in the frequency of spontaneous events likely results from receptor desensitization as a result of an accumulation of ACh in the extracellular space. In order to fully understand how synaptic-like nicotinic EPSPs arise in the dopaminergic axon, a better understanding of subcellular spatial dynamics of ACh release, diffusion in the extracellular space, and precise localization of AChE molecules around dopaminergic axons is needed.

Nicotinic signaling in axons from non-human primates

The role for striatal CIN activity in rewarding behaviors and neurodegenerative diseases has often been demonstrated through *in vivo* recordings in non-human primates (Kimura et al., 1984; Aosaki et al., 1994; Raz et al., 1996; Apicella et al., 1997). Yet, mechanistic research examining cholinergic modulation of DA release has largely been performed in rodent animal models. We found spontaneous and stimulated activation of nAChRs on dopaminergic axons of rhesus macaques that largely resembled axEPSPs recorded from mice in their kinetics, amplitude, and latency to onset. One interesting difference was in the stimulated axEPSP decay kinetics, which were significantly faster in macaque. This difference could result from a difference in mechanisms regulating axEPSP decay, such as AChE activity or channel properties, and would be predicted to result in a sharper integration time window in macaques for nAChR-evoked axonal APs. Therefore, the generation of axonal spikes may require stronger synchrony in cholinergic interneuron firing for primates than for rodents.

Independent input-output signaling of dopaminergic axons

The dendrites of DA neurons perform roles traditionally associated with axons, reliably backpropagating action potentials (Häusser et al., 1995; Hage and Khaliq, 2015) that cause the vesicular release of DA (Beckstead et al., 2004; Gantz et al., 2013; Gantz et al., 2018; Lebowitz et al., 2022; Hikima et al., 2021). In this study, we reveal properties of dopaminergic axons that are reminiscent of dendritic functions. We report the presence of spontaneously occurring EPSPs that have traits similar to those recorded in the somatodendritic region and are suggestive of integrative processing independently occurring in the axonal compartment. In this scenario, transmission at nAChRs is the excitatory input for integration. There is often also inhibitory input to balance out the excitatory drive. In the axon, this inhibition may come from axonal GABA-A receptors (Kramer et al., 2020). Interestingly, axonal GABA-A signaling lacks the rapid characteristics of axonal cholinergic transmission, perhaps setting a GABA-dependent permissive or restrictive tone for the generation of APs in the axon by cholinergic input.

Recent work has proposed that a mismatch exists between behaviorally relevant signals encoded by somatic action potential firing and DA release (Hamid et al., 2016; Mohebi et al., 2019; Liu et al., 2022). Furthermore, it has been shown that the activation of nicotinic receptors with carbachol puff or the synchronous activation of striatal CINs leads to striatal DA release (Threlfell et al., 2012; Cachepe et al., 2012) that is AP-mediated (Liu et al., 2022). Our data provide a detailed mechanism underlying this independent axonal signaling. We show that AP initiation in the dopaminergic axon occurs as a result of the spontaneous release of ACh. The spontaneous nature of these axonal APs strongly argues for their physiological relevance to an intact circuit *in vivo* and that they potentially contribute to a behavioral mismatch between somatic and axonal output. Our observation of cholinergic axEPSPs also reveals a mechanism for how dopaminergic axons function beyond the traditional presynaptic membrane roles in spike transmission to include postsynaptic membrane roles such as signal processing and integration.

We have taken the first look at endogenous subthreshold cholinergic transmission onto a central axon and found unexpected synaptic-like EPSPs. Presynaptic nAChRs are expressed throughout the brain, raising the possibility that other cell types exhibit axEPSPs that locally control neurotransmitter release through subthreshold modulation of excitability and the initiation of axonal action potentials.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
 - Lead contact
 - Materials availability
 - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
 - Mice

- Rhesus macaques
- **METHOD DETAILS**
 - Viral injections
 - Slicing and electrophysiology
 - Fluorescent imaging
 - Immunohistochemistry, clearing, confocal imaging, and neural reconstructions
- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.neuron.2022.07.011>.

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AUTHOR CONTRIBUTIONS

Conceptualization, P.F.K. and Z.M.K.; data curation, P.F.K., S.G.B.-W., and Z.M.K.; formal analysis, P.F.K., S.G.B.-W., and Z.M.K.; funding acquisition, P.F.K., B.B.A., A.H.N., and Z.M.K.; investigation, P.F.K., S.G.B.-W., A.C.C., G.A.C.-H., and M.A.G.E.; methodology, P.F.K., S.G.B.-W., A.C.C., R.Z., G.A.C.-H., A.H.N., M.A.G.E., and Z.M.K.; project administration, A.H.N., B.B.A., and Z.M.K.; resources, A.H.N., B.B.A., and Z.M.K.; software, P.F.K. and Z.M.K.; supervision, P.F.K., A.H.N., B.B.A., and Z.M.K.; validation, P.F.K., G.A.C.-H., A.H.N., and Z.M.K.; visualization, P.F.K. and Z.M.K.; writing—original draft, P.F.K. and Z.M.K.; writing—review & editing, P.F.K., S.G.B.-W., A.C.C., G.A.C.-H., A.H.N., M.A.G.E., B.B.A., and Z.M.K.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We worked to ensure sex balance in the selection of non-human subjects. One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in science. While citing references scientifically relevant for this work, we also actively worked to promote gender balance in our reference list.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Streptavidin cy5 conjugate	Invitrogen	SA1011
Bacterial and virus strains		
AAV9-CAG-Flex-TdTomato.WPRE.bGH	Penn vector core	AV-9-ALL864
AAV5-Ef1a-DIO-hchr2(H134R)-eYFP	UNC Vector Core	N/A
AAV9-hSyn-DIO-GRABACH3.0	Plasmid: (Jing et al., 2020) Virus: Vigene	Plasmid: Addgene, 121923
Chemicals, peptides, and recombinant proteins		
Acetylcholine	Sigma	A6625
Ambenonium	Tocris	0388
Atropine	Sigma	A0132
CGP55845	Tocris	1248
Conotoxin-P1a	Biomatic	N/A
D-AP5	Tocris	0106
dihydro-β-erythroidine hydrobromide (DHβE)	Tocris	2349
Gelatin from cold water fish skin	Sigma	G7041
Gramicidin	Sigma	G5002
Mecamylamine	Sigma	M9020
MFZ9-18	Amy Newman, PhD	N/A
NBQX	Tocris	1044
(+)-Sodium L-ascorbate	Sigma	A4034
Sodium Pyruvate	Sigma	P5280
SR95531 hydrobromide (Gabazine)	Tocris	1262
(±)-Sulpiride	Sigma	S8010
Tetrodotoxin (TTX)	Tocris	1078
Thiourea	Sigma	T7875
Urea	Sigma	U5128
Experimental models: Organisms/strains		
Ai9: Gt(ROSA)26Sor(tm9(CAG-tdTomato)Hze)	The Jackson Laboratory	RRID:IMSR_JAX:007909
ChAT-IRES-Cre::SV40pA::frr-neo-frr	The Jackson Laboratory	RRID:IMSR_JAX:006410
B6.SJL-Slc6a3(tm1.1(cre)Bkmm/J (DAT-IRES-cre)	The Jackson Laboratory	RRID:IMSR_JAX:006660

RESOURCE AVAILABILITY

Lead contact

The lead contact for this paper is Zayd M. Khaliq, Ph.D. (zayd.khaliq@nih.gov).

Materials availability

This study did not generate new unique reagents. Further information and requests for resources should be directed to and will be fulfilled by the lead contact.

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead author upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mice

All animal handling and procedures were approved by the animal care and use committee (ACUC) for the National Institute of Neurological Disorders and Stroke (NINDS) Intramural Research Program at the National Institutes of Health. Mice of both sexes were used throughout the study. Mice that underwent viral injections were injected at postnatal day 18 or older and were used for *ex vivo* electrophysiology and imaging 3–12 weeks after injection. The following strains were used: DAT-Cre (RRID:IMSR_JAX:006660); ChAT-cre (RRID:IMSR_JAX:006410); Ai9 (RRID:IMSR_JAX:007909).

Rhesus macaques

All experimental procedures were performed in accordance with the ILAR Guide for the Care and Use of Laboratory Animals and were approved by the Animal Care and Use Committee of the National Institute of Mental Health. Procedures adhered to applicable United States federal and local laws, regulations, and standards, including the Animal Welfare Act and Regulations (PL89–544; 1985 <https://www.nal.usda.gov/awic/animal-welfare-act>) and Public Health Service (PHS) Policy (PHS2002). Two female rhesus macaques (*Macaca mulatta*, *P* – 12.1yo, PAG lesion and rear split; *MJ* – 9.8yo, bilateral amygdala excitotoxic lesions) were used for this study. For brain extraction, the animals were sedated with ketamine/midazolam (ketamine 5–15 mg/kg, midazolam 0.05–0.3 mg/kg) and maintained on isoflurane. A deep level of anesthesia was verified by an absence of response to toe-pinch and absence of response to corneal reflex. The animal was perfused with ice-cold artificial CSF solution containing in mM: 124 NaCl; 23 NaHCO₃; 3 NaH₂PO₄; 5 KCl; 2 MgSO₄; 10 d-glucose; 2 CaCl₂.

METHOD DETAILS

Viral injections

All stereotaxic injections were conducted on a Stoelting QSI (Cat#53311). Mice were maintained under anesthesia for the duration of the injection and allowed to recover from anesthesia on a warmed pad. Viruses used for this study were: AAV9-CAG-Flex-TdTomato.WPRE.bGH (Penn Vector Core; AV-9-ALL864); AAV9-hSyn-DIO-GRABCh3.0 titer: $> 1 \times 10^{12}$, plasmid: Addgene (#121923), virus packaged by Vigene; and AAV5-Ef1a-DIO-hChR2(H134R)-eYFP, titer: 4×10^{12} , UNC vector core. Viral aliquots were injected (0.5–1 μ l) bilaterally into either the medial dorsal striatum (X: ± 1.7 Y: +0.8 Z: –3.3) or the SNc (X: ± 1.9 Y: –0.5 Z: –3.9) via a Hamilton syringe. At the end of the injection, the needle was raised at a rate of 0.1 to 0.2 mm per minute for 10 minutes before the needle was removed.

Slicing and electrophysiology

Brain slice experiments were performed on male and female adult mice of at least 6 weeks in age. Mice were anesthetized with isoflurane, decapitated, and brains rapidly extracted. Horizontal sections were cut at 330–400 μ m thickness on a vibratome while immersed in warmed, modified, slicing aCSF containing (in mM) 198 glycerol, 2.5 KCl, 1.2 NaH₂PO₄, 20 HEPES, 25 NaHCO₃, 10 glucose, 10 MgCl₂, 0.5 CaCl₂. Cut sections were promptly removed from the slicing chamber and incubated for 30–60 minutes in a heated (34°C) chamber with holding solution containing (in mM) 92 NaCl, 30 NaHCO₃, 1.2 NaH₂PO₄, 2.5 KCl, 35 glucose, 20 HEPES, 2 MgCl₂, 2 CaCl₂, 5 Na-ascorbate, 3 Na-pyruvate, and 2 thiourea. Slices were then stored at room temperature and used 30 min to 6 hours later. Following incubation, slices were moved to a heated (33–35°C) recording chamber that was continuously perfused with recording aCSF (in mM): 125 NaCl, 25 NaHCO₃, 1.25 NaH₂PO₄, 3.5 KCl, 10 glucose, 1 MgCl₂, 2 CaCl₂.

To prepare macaque brain slices, tissue blocks were prepared containing the caudate and putamen and were then mounted on a vibratome as described for mice. Coronal sections were cut at 300 μ m thickness in ice-cold sucrose cutting solution, filtered through a 0.22 μ m filter (Nalgene), that contained (in mM): 90 sucrose, 80 NaCl, 3.5 KCl, 24 NaHCO₃, 1.25 NaH₂PO₄, 4.5 MgCl₂, 0.5 MgCl₂, 10 glucose. Cut sections were promptly placed into a heated (34°C) chamber with the same sucrose solution and incubated in the warm bath for 30–60 minutes. Slices were then stored at room temperature and used 30 minutes to 3 days later. The incubation solution was exchanged for clean solution every 12 hours.

Perforated-patch recordings were made using borosilicate pipettes (5–10 M Ω) filled with internal solution containing (in mM) 135 KCl, 10 NaCl, 2 MgCl₂, 10 HEPES, 0.5 EGTA, 0.1 CaCl₂, adjusted to a pH value of 7.43 with KOH, 278 mOsm. Pipette tips were back filled with ~ 1 μ l of clean internal. Pipettes were then filled with internal containing between 80 and 100 μ g/mL gramicidin. Patch integrity was monitored by the addition of either Alexa-488 (0.01 mM, axons filled with tdTomato) or Alexa-594 (0.01 mM, axons labeled with MFZ9–18) to the gramicidin-containing internal. Experiments where the perforation was lost were included up until the point where the recording substantially changed in quality (depolarized to above –45 mV, or changed input resistance by more than 15%). To enable post-hoc reconstruction, pipette solutions in some experiments included 0.1–0.3% w/v neurobiotin (Vector Labs). Current clamp recordings were manually bridge balanced. Conotoxin-P1a was synthesized based on the published structure by Biomatik (Ontario, CA). To isolate ACh transmission, we used isolation solutions (isolation aCSF) containing CNQX (10 μ M), AP-5 (20 μ M), CGP (300 nM), sulpiride (300 nM), atropine (30 nM), and gabazine (10 μ M) to block AMPA, NMDA, GABA-B, dopamine D2, muscarinic, and GABA-A receptors.

Electrical stimulation was evoked with tungsten bipolar electrodes (150 μm tip separation, MicroProbes). For experiments where the site of electrical stimulation is distal to the site of imaging or recording, electrodes were placed at the caudal end of horizontal brain slices. Stimulations were evoked using an Isoflex (A.M.P.I.), amplitudes ranging from 0.8 to 20 V. The slices were allowed to acclimate in the recording chamber for at least 15 minutes before experiments began for all conditions, including for experiments where the isolation solution was added to the background aCSF to ensure receptor antagonist equilibrium had been achieved.

Fluorescent imaging

A white light LED (Thorlabs; SOLIS-3C) was used in combination with a tdTomato (Thorlabs; TLV-U-MF2-TOM) or EGFP (Chroma; 49002) filter set to visualize the dopaminergic axons (tdTomato) or for use with the ChR2 and GRAB_{ACh} (GFP). For visualizing the GRAB_{ACh} signals, a photodiode (New Focus) was mounted on the top port of the Olympus BX-51WI. For opsin activation, the LED was controlled with a TTL pulse (2-5 ms).

For visualization of dopaminergic axons in brain slices containing rhesus macaque tissue, a brain slice was moved to a separate incubation chamber with the regular sucrose solution, with the addition of 100 nM MFZ9-18 and saturating sulpiride (10 μM) for 10 to 60 minutes. Slices were then moved to a recording chamber and axons visualized with blue light (EGFP filter set). MFZ9-18 was newly synthesized by Gisela Andrea Camacho-Hernandez, Ph.D. in the lab of Amy Newman, Ph.D. according to previously published protocols (Eriksen et al., 2009).

Immunohistochemistry, clearing, confocal imaging, and neural reconstructions

After electrophysiology or imaging, slices with neurobiotin filled axons were fixed overnight in 4% paraformaldehyde (PFA) in phosphate buffer (PB, 0.1M, pH 7.6). Slices were subsequently stored in PB until immunostaining and cleared using a modified CUBIC protocol, chosen because it does not quench endogenous fluorescence (Susaki et al., 2015). For the immunostaining and CUBIC clearing, all steps were performed at room temperature on a shaker plate. Slices were placed in CUBIC reagent 1 for 1-2 days, washed in PB 3 x 1 hour each, placed in blocking solution (0.5% fish gelatin (Sigma) in PB) for 3 hours. Slices were directly placed in streptavidin-Cy5 conjugate at a concentration of 1:1000 in PB for 2-3 days. Slices were washed 3 times for 2 hours each and were then placed in CUBIC reagent 2 overnight. Slices were mounted on slides in reagent 2 in frame-seal incubation chambers (Bio-Rad SLF0601) and coverslipped (#2 glass). Slices were imaged through 20 $\times$, 0.8 nA and 5 $\times$, 0.3 nA objectives on an LSM 800 confocal microscope (Zeiss) and taken as tiled z-stacks using Zen Blue software in the NINDS light imaging facility. Striatal axons were reconstructed using Neurolucida (MBF bioscience).

QUANTIFICATION AND STATISTICAL ANALYSIS

Fast voltages recorded from small compartments such as axons can appear distorted, especially when the recording is performed with perforated-patch which puts limitations on lowering series resistance (Oláh et al., 2021). For this reason, putative spontaneous APs recorded for the analysis in Figure 4 were compared to known APs stimulated in striatal perforated-patch recordings and to soma-generated spontaneous APs recorded in perforated-patch from the main trunk of the dopaminergic axon (Kramer et al., 2020). First, recordings were differentiated and signals above 10 standard deviations of the mean were selected as likely APs. A weighted z-score method was then used to establish which putative spontaneous APs to exclude from analysis that considered the absolute positive (P) and negative (N) peak dV/dt values, as well as the ratio $R = \frac{P}{|N|}$ and the time $T = \tau(N) - \tau(P)$, between the two peak values. The time difference T was also used as a measurement of the AP half-width. These four values were established for the reference APs, and the putative spontaneous APs were compared against the averages using a z-score.

$$Z_x = \frac{(x - \bar{x})}{\sigma}$$

The z-scores were then combined using a weighted sum of squares to give weight to the relative values, which are more reliable to compare between experiments when doing axonal perforated-patch recordings (Oláh et al., 2021).

$$\text{SumSq} = .5(x_P - \bar{x})^2 + .5(x_N - \bar{x})^2 + (x_R - \bar{x})^2 + 2(x_T - \bar{x})^2$$

Putative APs with SumSq values larger than 21 (z-score of 3 for P and N , z-score of 2 for R and T) were excluded.

Spontaneous axEPSPs were detected using Neuromatic (Rothman and Silver, 2018), written for use with Igor Pro (Wavemetrics). We used event detection in Neuromatic that is based on a simple level detection algorithm (Kudoh and Taguchi, 2002). Levels were set at least 3 times the amplitude of baseline recording noise (between 0.3 and 1 mV), a sliding baseline average (5 ms) was used along with onset (1 SD, 30 ms) and peak detection (1 to 2 SD, 60 ms) parameters. After automated detection, traces were scanned for false positives, as defined by events without a clear decay following a rapid rise, and any such false positives were eliminated. Undetected events were only rarely manually added when automatic detection did not label a clear and obvious event, this was in order to not bias the data. See Figure S1 for three examples of automated peak detection output from three different axonal recordings. Spontaneous axEPSP kinetics were analyzed using built-in features in Neuromatic. Rise times were calculated as the time from 10 to 90% of the peak axEPSP amplitude. Decay was calculated as the time to 36.7% of the peak amplitude (1 time

constant). Half-width was calculated as the time between the rise and decay points at half the peak axEPSP amplitude. Latency to onset was calculated as time between stimulus onset and 5% of peak axEPSP amplitude.

Analysis was conducted in Igor Pro and Prism 8 (GraphPad). Data in text are reported in terms of means for parametric data and medians for non-parametric data. For parametric data, data in the text are expressed as means, t-tests were used for two-group comparison, and ANOVA tests were used for more than two group comparisons, followed by a Bonferonni or Tukey post-hoc test for analysis of multiple comparisons. Parametric data are shown as averages $\pm$ standard error of the mean. For non-parametric data sets, data in the text are expressed as medians, Mann-Whitney U tests were used to compare two groups without repeated measures and a Wilcoxon test was used to compare two groups with repeated measures. A Kruskal-Wallis test was used to compare more than two groups followed by Dunn's multiple comparison test for between-subject comparisons, while a Friedman test was used for comparisons within group (repeated measures) and a Dunn's test used for multiple comparisons. Error bars on cumulative distribution plots are $\pm$ standard error of the mean. Non-parametric data are shown with medians, and box-and-whisker plots are shown with the box going from the 25th to 75th percentiles, and the whiskers going from the minimum to maximum values.